

DD_FUL-08 - Anchor Date Change

Version: v1.0 **Bundle:** 06_fulfillment **Status:** Draft for review **Primary owner:** Fulfillment / subscription workflow team
Related DDs: DD_SUB-LM-01, DD_SUB-WF-FRAMEWORK-01, DD_BIL-01, DD_BIL-02-GEN-01, DD_BIL-03, DD_BIL-CFG-01, DD_BIL-02-ADJ-01, DD_EM-CFG-04, FOUNDATION_DROOLS

1. Purpose

FUL-08 defines how SOPHIX changes a subscription's billing anchor date.

The billing anchor date is the day used to compute cycle windows, recurring charge timing, invoice period boundaries, prepaid wallet cycle debit timing, dunning expectations, and customer communication dates.

Use FUL-08 when:

- customer asks to move billing date from the 1st to the 15th;
- operator aligns anchor date after relocation, corporate account change, or manual correction;
- care agent needs to correct a wrong anchor date created during onboarding;
- billing team needs an auditable operation before BIL-03 cycle close uses the new date.

FUL-08 is not the billing engine. It orchestrates the change, validates policy, asks Billing to preview/compute financial impact, obtains approval where needed, updates subscription lifecycle state through SUB-LM-01, and emits events for Billing and reporting.

2. Scope And Boundaries

Capability	Owner
Anchor change request, validation, approval, operation audit	FUL-08
Subscription master anchor field update	SUB-LM-01
Billing impact preview and proration charge/credit	BIL-01 / BIL-02-ADJ
Cycle close use of new anchor date	BIL-03
Billable event codes for anchor change fee/proration	BIL-CFG-01
Configurable approval policy	EM-CFG-04
Operator-variable eligibility/fee/approval policy	Drools

Boundary rules:

- FUL-08 must not directly generate invoices.
- FUL-08 must not directly edit Billing ledger, invoice, wallet, or adjustment rows.
- FUL-08 must not directly update subscription DB fields if SUB-LM-01 exposes an owning API/service; call the owner.
- EM-CFG-04 owns approval tasks and policy. FUL-08 stores approval references and applies final outcome.
- Camunda is used for the anchor-change process itself if the operator wants async approval/payment/timer orchestration. EM-CFG-04 starts Camunda only for approval policies in BPMN mode.

3. Process Overview

```
flowchart LR
    A["User requests anchor date change"] --> B["Validate subscription and requested day"]
    B --> C["Run Drools policy"]
    C --> D["Call BIL preview"]
    D --> E["Approval required?"]
    E -->|Yes| F["EM-CFG-04 approval"]
    E -->|No| G["Apply"]
    F --> G
    G --> H["SUB-LM updates anchor date"]
    G --> I["BIL applies fee/proration if any"]
    H --> J["Emit AnchorDateChanged"]
```

4. Status Model

Status	Meaning
DRAFT	Request captured but not submitted.
VALIDATING	Validation and billing preview running.
PENDING_APPROVAL	EM-CFG-04 approval is open.
PENDING_PAYMENT	Customer must pay anchor-change fee before apply.
APPROVED	Ready to apply.
APPLIED	Anchor change completed.
REJECTED	Policy or approval rejected.
CANCELLED	Request cancelled before apply.
FAILED	Apply failed and needs retry/manual repair.

5. Data Model

5.1 anchor_date_change_request

What it is for: stores the business request to change a subscription billing anchor date.

When to use it: every customer, care, billing, or admin anchor-date change must create this row before subscription state changes.

Sample meaning: adc-001 requests moving subscription SUB-700045 from anchor day 1 to anchor day 15, effective next cycle.

Column	Type	Purpose	Sample
request_id	TEXT PK	Anchor change request id.	adc-001
operator_code	TEXT	Tenant/operator.	WIK
subscription_id	TEXT	Subscription being changed.	SUB-700045
account_id	TEXT NULL	Account id.	ACC-900045
customer_id	TEXT NULL	Customer id.	CUS-100045
current_anchor_day	INT	Current anchor day.	1
requested_anchor_day	INT	Requested day of month, 1-28 in v1.0.	15
effective_mode	TEXT	IMMEDIATE, NEXT_CYCLE, SCHEDULED_AT.	NEXT_CYCLE
scheduled_effective_at	TIMESTAMP NULL	Effective date if scheduled.	2026-07-15T00:00:00Z
reason_code	TEXT	Business reason.	CUSTOMER_REQUEST
source_channel	TEXT	BACKOFFICE, SELF_CARE, API, BATCH.	BACKOFFICE
status	TEXT	Request status.	PENDING_APPROVAL
requested_by_user_id	TEXT	Requester.	usr-care-001
approval_request_id	TEXT NULL	EM-CFG-04 approval request id.	apr-adc-001
approval_mode	TEXT NULL	EM-CFG-04 mode.	MULTI_STEP
camunda_process_instance_id	TEXT NULL	FUL-08 process instance if workflow used.	cam-adc-001
created_at	TIMESTAMP	Creation timestamp.	2026-06-20T09:00:00Z
applied_at	TIMESTAMP NULL	Apply timestamp.	2026-07-01T00:03:00Z

Recommended indexes:

- (operator_code, subscription_id, status).
- (operator_code, status, created_at DESC).

- (approval_request_id).

5.2 anchor_date_change_preview

What it is for: stores the billing preview returned before applying the change.

When to use it: every submitted request must have a preview so the customer/operator can see fee, credit, proration, next cycle window, and payment requirement.

Sample meaning: adcp-001 says moving to anchor day 15 creates KES 1400 prorated charge and no approval fee.

Column	Type	Purpose	Sample
preview_id	TEXT PK	Preview id.	adcp-001
request_id	TEXT	Anchor change request.	adc-001
current_cycle_start	DATE	Current cycle start.	2026-06-01
current_cycle_end	DATE	Current cycle end.	2026-06-30
new_cycle_start	DATE	New cycle start.	2026-07-15
new_cycle_end	DATE	New cycle end.	2026-08-14
financial_impact_type	TEXT	NONE, CHARGE, CREDIT, MIXED.	CHARGE
proration_amount	NUMERIC(14,2)	Charge/credit estimate.	1400.00
anchor_change_fee	NUMERIC(14,2)	Optional fee.	0.00
currency_code	TEXT	Currency.	KES
payment_required	BOOLEAN	Whether customer must pay before apply.	true
billing_preview_ref	TEXT	BIL preview id/reference.	bil-prev-adc-001
created_at	TIMESTAMP	Creation timestamp.	2026-06-20T09:01:00Z

Recommended indexes:

- (request_id).
- (billing_preview_ref).

5.3 anchor_date_change_apply_log

What it is for: records the concrete apply attempts and downstream calls.

When to use it: every apply/retry writes an attempt row, including SUB-LM and Billing call references.

Sample meaning: adcal-001 records that SUB-LM update succeeded and BIL-01 proration event was accepted.

Column	Type	Purpose	Sample
apply_log_id	TEXT PK	Apply attempt id.	adcal-001
request_id	TEXT	Anchor change request.	adc-001
attempt_no	INT	Attempt number.	1
status	TEXT	STARTED, SUB_UPDATED, BILLING_APPLIED, COMPLETED, FAILED.	COMPLETED
sub_lm_request_ref	TEXT NULL	SUB-LM call reference.	sub-anchor-adc-001
billing_apply_ref	TEXT NULL	BIL apply reference.	bil-apply-adc-001
failure_code	TEXT NULL	Failure code.	null
failure_detail	TEXT NULL	Failure details.	null
started_at	TIMESTAMP	Attempt start.	2026-07-01T00:00:00Z
completed_at	TIMESTAMP NULL	Attempt completion.	2026-07-01T00:03:00Z

Recommended indexes:

- (request_id, attempt_no).
- (status, started_at DESC).

6. APIs

6.1 Preview Anchor Change

POST /api/subscriptions/{subscription_id}/anchor-date-change/preview
Content-Type: application/json

```
{
  "operatorCode": "WIK",
  "requestedAnchorDay": 15,
  "effectiveMode": "NEXT_CYCLE",
  "reasonCode": "CUSTOMER_REQUEST",
  "sourceChannel": "BACKOFFICE"
}
```

FUL-08 validates subscription state, runs Drools policy, then calls Billing preview.

6.2 Create Anchor Change Request

POST /api/subscriptions/{subscription_id}/anchor-date-change
Idempotency-Key: adc-sub-700045-202607-anchor15
Content-Type: application/json

```
{
  "operatorCode": "WIK",
  "requestedAnchorDay": 15,
  "effectiveMode": "NEXT_CYCLE",
  "reasonCode": "CUSTOMER_REQUEST",
  "sourceChannel": "BACKOFFICE",
  "confirmedPreviewRef": "bil-prev-adc-001"
}
```

6.3 Approval Evaluation

FUL-08 calls EM-CFG-04 when rule/policy requires approval:

POST /api/approvals/evaluate
Content-Type: application/json

```
{
  "operatorCode": "WIK",
  "moduleCode": "FUL-08",
  "actionCode": "ANCHOR_DATE_CHANGE_APPROVAL",
  "subjectType": "ANCHOR_DATE_CHANGE",
  "subjectId": "adc-001",
  "context": {
    "subscriptionId": "SUB-700045",
    "currentAnchorDay": 1,
    "requestedAnchorDay": 15,
    "financialImpactType": "CHARGE",
    "prorationAmount": 1400.00,
    "currencyCode": "KES",
    "sourceChannel": "BACKOFFICE"
  }
}
```

6.4 Approval Outcome Callback

POST /internal/anchor-date-change/{request_id}/approval-outcome
Content-Type: application/json

```
{
  "approvalRequestId": "apr-adc-001",
  "outcome": "APPROVED",
  "decisionComment": "Approved customer billing date alignment."
}
```

6.5 Apply Anchor Change

POST /api/anchor-date-change/{request_id}/apply
Idempotency-Key: apply-adc-001
Content-Type: application/json

Apply behavior:

1. call SUB-LM-01 to update anchor day/effective date;
2. call Billing to apply fee/proration if preview requires it;
3. emit AnchorDateChanged.

7. Business Rules

Rule	Description	Reason
R-FUL-08-01	Subscription must be active or in an allowed state.	Avoid changing closed subscriptions.
R-FUL-08-02	Anchor day must be between 1 and 28 in v1.0.	Avoid month-length edge cases.
R-FUL-08-03	Only one pending anchor change per subscription.	Prevent conflicting cycle windows.
R-FUL-08-04	Billing preview is required before request confirmation.	Customer/finance transparency.
R-FUL-08-05	Approval must go through EM-CFG-04 when policy requires it.	Governance.
R-FUL-08-06	Customer payment must clear before apply if Billing marks paymentRequired=true.	Revenue protection.
R-FUL-08-07	Apply must be idempotent.	Prevent double proration.
R-FUL-08-08	BIL-03 must use the new anchor only after successful apply event/state.	Cycle consistency.

8. Rule Engine Guidance

Use Drools for configurable policy:

Rule file	Purpose
anchor-date-change-eligibility.drl	Decide if requester/source/subscription may change anchor.
anchor-date-change-approval.drl	Decide whether EM-CFG-04 approval is required.
anchor-date-change-payment-policy.drl	Decide payment-first behavior where operator policy differs from Billing default.

Example:

```
package com.sophix.fulfillment.anchor

rule "Backoffice anchor change with large proration requires approval"
when
    $f : AnchorChangeFact(sourceChannel == "BACKOFFICE", prorationAmount >= 1000.00)
    $d : AnchorChangeDecision()
then
    $d.setApprovalRequired(true);
    $d.setApprovalPolicyCode("ANCHOR_CHANGE_SUPERVISOR");
end
```

9. Events

Event	Topic	When emitted
AnchorDateChangeRequested	sophix.fulfillment.anchor-change.requested	Request created.
AnchorDateChangeApprovalRequired	sophix.fulfillment.anchor-change.approval-required	Approval started.
AnchorDateChanged	sophix.fulfillment.anchor-change.changed	Anchor applied.
AnchorDateChangeRejected	sophix.fulfillment.anchor-change.rejected	Policy/approval rejected.
AnchorDateChangeFailed	sophix.fulfillment.anchor-change.failed	Apply failed.

10. Frontend/API Consumers

Area	APIs/modules used
Backoffice customer care	Preview/create/apply FUL-08 APIs.

Area	APIs/modules used
Approval inbox	EM-CFG-04 task APIs.
Billing views	BIL preview/apply refs from FUL-08 request detail.
Self-care	Preview/create only if operator policy allows.
Reporting	FUL-08 events and REP-01.

11. Infrastructure Recommendations

Run FUL-08 inside existing fulfillment/subscription workflow API and worker deployments.

Component	Recommendation
API	Existing subscription/fulfillment API exposes anchor-date endpoints.
Database	Fulfillment/subscription workflow schema.
Workers	Shared workflow worker for apply retry, outbox, payment wait.
Camunda	Optional for full async flow; use if approval/payment/scheduled apply needs orchestration.
Drools	Existing KIE/Drools foundation.
Billing integration	Service client to BIL-01/BIL-02/BIL-03 APIs/events.

AWS notes: use existing ECS/EKS workflow services, RDS PostgreSQL, MSK/Kafka, Secrets Manager, and CloudWatch/Prometheus alerts for stuck requests and failed apply retries.

12. Acceptance Criteria

- Anchor change request/preview/apply tables exist.
- Preview calls Billing and stores financial impact.
- Approval uses EM-CFG-04 when policy requires.
- Apply updates subscription anchor through SUB-LM-01 and financial impact through Billing APIs.
- Apply is idempotent and retryable.
- BIL-03 can consume/apply the new anchor after successful event/state.
- Backoffice can preview, submit, approve, and apply anchor change.